

HYDROGENATED NBR (HNBR) PATENT LANDSCAPE: LOW-VISCOSITY, LOW-UNSATURATION, AND CATALYST-CONTROLLED SYNTHESIS ROUTES

1. ABSTRACT

This report synthesizes four primary patents addressing hydrogenated nitrile rubber (HNBR) with emphasis on low residual unsaturation, reduced molecular weight, and improved processability. Across the set, the target material is consistently defined as hydrogenated nitrile rubber derived from nitrile rubber/NBR precursors containing conjugated diene and α,β -unsaturated nitrile repeat units, with the desired outcome being a saturated or reduced-unsaturation elastomer exhibiting improved heat, ozone, oxidative, and aging resistance. The patents fall into two main technical families. US7772328B2 and EP2473280B1 focus on molecular-weight reduction and/or simultaneous hydrogenation-metathesis using ruthenium/osmium metathesis catalysts, often with co-olefins, to generate lower-Mw HNBR with narrower molecular-weight distributions and lower Mooney viscosity. US7772328B2 specifically demonstrates controlled depolymerization/metathesis of HNBR feedstocks using ethylene or 1-hexene under elevated pressure and temperature, while EP2473280B1 and US9150669B2 disclose one-step simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen, using Ru- or Os-based catalysts to achieve low RDB and reduced Mw. EP3827029A1 represents a different but complementary manufacturing trend: it centers on hydrogenating NBR in solution with Ru, Pd, or Rh catalysts at controlled catalyst loadings, temperature, pressure, and time, while minimizing catalyst residues that impair color, aging, and gel content. Overall, the landscape shows a strong move toward catalyst-engineered, solution-phase processes that combine hydrogenation with chain scission or metathesis to tune viscosity and molecular-weight distribution, alongside downstream catalyst-removal strategies to preserve HNBR quality. The evidence suggests that practical HNBR synthesis benefits from moderate-to-high hydrogenation pressure, temperatures around 60-200°C depending on catalyst system, and careful control of catalyst loading and co-olefin choice to balance hydrogenation rate, residual unsaturation, and polymer integrity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a low-molecular-weight HNBR preparation route based on olefin metathesis of hydrogenated nitrile rubber feedstocks. The polymer is dissolved in monochlorobenzene and contacted with a co-olefin such as ethylene or 1-hexene under agitation, with a metathesis catalyst used to reduce molecular weight and narrow the molecular-weight distribution. - Rubber concentration: 15 wt.-% solid — Example 1 - Rubber concentration: 15 wt.-% solid — Example 2 - Rubber concentration: 6 wt.-% solid — Example 3 - Rubber concentration: 6 wt.-% solid — Example 4 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - Mw: 85 kg/mol — starting polymer - Mw: 296 kg/mol — starting polymer - PDI: 3.4 — starting substrate feedstock - PDI: 2.0 — metathesized product - PDI: 1.90 — Table 2 - Mooney viscosity (ML 1+4 @100°C): 73 — starting polymer - Mooney viscosity (ML 1+4 @100°C): 43 — Experiment 1 - Mooney viscosity (ML 1+4 @100°C): 28 — Experiment 2 - Mooney viscosity (ML 1+4 @100°C): 3 — Experiment 3 - Mooney viscosity (ML 1+4 @100°C): 31 — Experiment 4 - Mw: 189000 g/mol — Experiment 1 - Mw: 136000 g/mol — Experiment 2 - Mw: 59000 g/mol — Experiment 3 - Mw: 111000 g/mol — Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-treated HNBR cement at 15 wt.-% solids was processed under 6.9×10^5 Pa to demonstrate metathesis-based molecular-weight reduction. - Example 2: Repeated ethylene metathesis at the same pressure showed further control of molecular-weight reduction relative to the starting polymer. - Example 3: Lower-solids cement (6 wt.-%) again under ethylene pressure produced a stronger reduction in Mw and PDI. - Example 4: Use of 1-hexene at 80° C. demonstrated that co-olefin identity affects the extent of depolymerization and final viscosity. - Example 4: The table data showed a substantial narrowing of PDI to 1.90 and Mooney viscosity reduction to 3 in the most aggressive condition.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR, a key objective for improving processability while retaining HNBR performance. It also demonstrates that metathesis/co-olefin chemistry can be used to tune Mw and Mooney viscosity after hydrogenation.

Patent 2

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of partially unsaturated NBR using ruthenium, palladium, or rhodium compounds, with emphasis on controlling catalyst residue and product aging behavior. The process is designed to produce HNBR with good aging properties while avoiding the costly and complicated post-hydrogenation catalyst-removal steps common in the prior art. - Ruthenium content: 10 ppm to 200 ppm — Claim 1 - Ruthenium content: 10 ppm to 150 ppm — Claim 1 - Ruthenium content: 10 ppm to 120 ppm — Claim 1 - Ruthenium content: 10 ppm to 79 ppm — Claim 1 - Ruthenium content: 42 ppm to 79 ppm — Claim 1 - Palladium content: 20 ppm to <67 ppm — Claim 1 - Palladium content: 44 ppm to <67 ppm — Claim 1 - Rhodium content: 50 ppm to <270 ppm — Claim 1 - Rhodium content: 79 ppm to <270 ppm — Claim 1 - Rhodium content: 50 ppm to 250 ppm — Claim 1 - Rhodium content: 50 ppm to 170 ppm — Claim 1 - Rhodium content: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700 000 pascals to 15 000 000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Gel content increase during aging: less than 10% — Description - Mooney viscosity increase during aging: less than 40% — Description - Rhodium content: 58-60 ppm — cited prior art EP-A-0482391 - Pd catalyst loading: 0.45 parts — cited prior art EP-A-1454924 - Rh removal rate: 99% — cited prior art CN-A-101704926 - Purified HNBR rhodium content: 5.8 to 9.2 ppm — cited prior art EP-A-0431370

Relevant Experimental Evidence - Example data were not detected by the parser; the patent content is primarily claim- and process-focused. - The disclosure emphasizes that lower catalyst residue improves aging, gel content, and color stability. - The patent contrasts homogeneous and heterogeneous catalyst recovery methods, including filtration, resin beds, ultrafiltration, ion-exchange resins, and ionic liquids. - The claimed process window defines catalyst loading and reaction conditions for Ru, Pd, and Rh systems. - The invention positions catalyst minimization and retention of product quality as the central manufacturing advance.

Technical Relevance This patent is relevant because it addresses a major industrial issue in HNBR synthesis: how to hydrogenate NBR efficiently while minimizing residual noble-metal contamination. The disclosed catalyst and process windows are directly applicable to producing cleaner, more stable HNBR.

Patent 3

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process in which nitrile rubber is simultaneously hydrogenated and metathesized in the presence of hydrogen, optionally with a co-olefin, using Ru- or Os-based catalysts. The approach is intended to produce HNBR with lower molecular weight, lower Mooney viscosity, and narrower molecular-weight distribution than conventional hydrogenation routes. - Catalyst amount: 0,057 phr — Example 1 - Catalyst amount: 0,080 phr — Example 1 - Catalyst amount: 0,06 phr — Example 1 - Catalyst amount: 0,055 phr — Example 1 - Metal content: 65,6 ppm — Example 1 - Metal content: 91,3 ppm — Example 1 - Metal content: 65,6 ppm — Example 1 - Metal content: 65,6 ppm — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% — Example 1 - RDB: 2,0% — Example 1 - RDB: 0,7% — Example 1 - RDB: 15,4% — Example 1 - RDB: 2,9% — Example 1 - RDB: 0,4% — Example 1 - RDB: 0,5% — Example 1 - RDB: 3,1% — Example 1 - RDB: 0,6% — Example 1 - RDB: 0,3% — Example 1 - RDB: 1,3% — Example 1 - Mw: 191541 g/mol — Example 1 - Mn: 68643 g/mol — Example 1 - PDI: 2,79 — Example 1 - Mooney viscosity: 38,2 — Example 1 - Mw: 135772 g/mol — Example 1 - Mn: 58259 g/mol — Example 1 - PDI: 2,33 — Example 1 - Mooney viscosity: 20,2 — Example 1 - Mw: 195304 g/mol — Example 1 - Mn: 64130 g/mol — Example 1 - PDI: 3,05 — Example 1 - Mooney viscosity: 69,3 — Example 1 - Mw: 144212 g/mol — Example 1 - Mn: 59851 g/mol — Example 1 - PDI: 2,45 — Example 1 - Mooney viscosity: 32,6 — Example 1

Relevant Experimental Evidence - Example 1: Compared Grubbs III, Wilkinson's, and Grubbs II catalysts at similar metal contents and showed different hydrogenation rates. - Example 1: Grubbs III gave faster hydrogenation in the first 120 min than Grubbs II or Wilkinson's catalyst. - Example 1: The catalyst amount and metal content were correlated with distinct RDB decay profiles. - Example 1: Final HNBR properties showed that catalyst choice strongly affected Mw, PDI, and Mooney viscosity. - Example 1: The data support simultaneous hydrogenation-metathesis as a route to low-viscosity HNBR.

Technical Relevance This patent is directly relevant because it teaches a simultaneous hydrogenation/metathesis route to HNBR with reduced molecular weight and low residual unsaturation. It also provides comparative catalyst data that help identify effective catalyst systems for tuning HNBR properties.

Patent 4

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight HNBR by contacting nitrile rubber with a ruthenium- or osmium-based metathesis catalyst in the presence of hydrogen and optionally a co-olefin. The process is designed to combine hydrogenation and metathesis in a single step, yielding HNBR with low residual double bond content, reduced Mw, and narrower molecular-weight distribution. - Catalyst amount: 0.041 phr — Example 1 - Catalyst amount: 0.045 phr — Example 1 - Catalyst amount: 0.060 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: NIL — Example 1 - Co-catalyst: NIL — Example 1 - Co-catalyst: Triphenylphosphine — Example 1 - Co-catalyst amount: 1.0 phr — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% — Example 1 - RDB: 2.5% — Example 1 - RDB: 7.0% — Example 1 - RDB: 15.4% — Example 1 - RDB: 0.6% — Example 1 - RDB: 4.5% — Example 1 - RDB: 3.1% — Example 1 - RDB: 0.5% — Example 1 - RDB: 3.9% — Example 1 - RDB: 1.3% — Example 1 - RDB: 0.4% — Example 1 - RDB: 3.5% — Example 1 - RDB: 0.7% — Example 1 - Mw: 113371 g/mol — Example 1 - Mn: 48352 g/mol — Example 1 - PDI: 2.34 — Example 1 - Mooney viscosity: 33.3 — Example 1 - Mw: 133758 g/mol — Example 1 - Mn: 54595 g/mol — Example 1 - PDI: 2.45 — Example 1 - Mooney viscosity: 41.3 — Example 1 - Mw: 195304 g/mol — Example 1 - Mn: 64130 g/mol — Example 1 - PDI: 3.05 — Example 1 - Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared at similar metal contents. - Example 1: The catalyst according to the invention showed more efficient hydrogenation than the industrial norm catalyst. - Example 1: The data showed that hydrogenation with the inventive catalyst reduced molecular weight of the nitrile polymer. - Example 1: Triphenylphosphine co-catalysis was used with Wilkinson's catalyst and affected the hydrogenation profile. - Example 1: Final HNBR properties demonstrated lower Mw and lower Mooney viscosity for the inventive catalyst system.

Technical Relevance This patent is relevant because it provides a catalyst-system approach for producing low-molecular-weight HNBR in a single step. The comparative example data are useful for selecting catalyst/co-catalyst combinations that improve hydrogenation efficiency and reduce viscosity.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7772328B2 and EP2473280B1/US9150669B2 all pursue lower Mw and lower Mooney viscosity, but US7772328B2 does so mainly by metathesis/depolymerization of HNBR feedstock, whereas EP2473280B1 and US9150669B2 integrate hydrogenation with metathesis in one step.
- EP2473280B1 and US9150669B2 both use Ru/Os metathesis-type catalyst families in hydrogen, with optional co-olefin, and both report final Mw values in the ~113,000-195,000 g/mol range and PDI values around 2.3-3.1 in the examples.
- US7772328B2 uses ethylene at 6.9×10^5 Pa and 1-hexene at 80° C. to drive chain scission/metathesis, while EP2473280B1 and US9150669B2 emphasize catalyst identity and hydrogenation profile rather than a single co-olefin condition.
- EP3827029A1 differs from the other three by focusing on conventional solution hydrogenation of NBR with Ru/Pd/Rh catalysts and on catalyst residue control, rather than on metathesis-driven molecular-weight reduction.
- Catalyst loading is lowest in the metathesis/hydrogenation patents (about 0.041-0.080 phr in the examples) and higher catalyst-residue windows are explicitly claimed in EP3827029A1 in ppm terms for Ru, Pd, and Rh.
- Temperature and pressure are explicitly claimed only in EP3827029A1 (60-200°C, 700,000-15,000,000 Pa), while the metathesis-based patents provide example temperatures such as 80° C. and focus more on catalyst system performance and RDB decay.

4. CONCLUSION

The patent set shows two complementary routes to HNBR: conventional solution hydrogenation with careful catalyst-residue management, and simultaneous hydrogenation-metathesis for direct molecular-weight reduction. For practical HNBR synthesis, the strongest evidence supports using Ru-based catalyst systems in solution, with hydrogenation conditions around 60-200°C and elevated hydrogen pressure when conventional hydrogenation is used, or around 80°C with low phr catalyst loadings when metathesis-assisted chain scission is desired. The most effective route for low-viscosity HNBR appears to be simultaneous hydrogenation/metathesis with controlled co-olefin selection, because it can reduce Mw, narrow PDI, and lower Mooney viscosity in a single step. Where product color, aging, and gel stability are critical, catalyst residue minimization and post-reaction purification remain important. Overall, the evidence favors low catalyst loading, Ru-based single-site catalysts, and process control that balances residual unsaturation reduction against over-degradation of the polymer backbone.

5. REFERENCES

- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>